

MAT2040 Linear Algebra

Midterm Exam
SSE, CUHK(SZ)

31 Oct 2020

Seat No.: _____ Student ID: _____

- Answer each question in the space after the question.
- For questions in Part I, you can give answers without explanations.
- For questions in Part II, be sure to give **full explanations** for your answers. The solution only is not supposed to be a complete answer.

Question	Points	Score
1	5	
2	6	
3	6	
4	10	
5	7	
6	12	

Question	Points	Score
7	10	
8	10	
9	10	
10	12	
11	12	
Total:	100	

This page has no questions, but a table of notations.

Table of Notations

$\mathbb{R}$	set of real numbers
$\mathbb{R}^n$	Without otherwise specified, all matrices have entries from $\mathbb{R}$ set of all (column) vectors of n entries from $\mathbb{R}$ n -dimensional Euclidean vector spaces
$0, 1, \dots$	scalar values
$\mathbf{0}$	zero vector in $\mathbb{R}^n$, where n is implied in the context
I, I_n	the $n \times n$ identity matrix
A^T	transpose of matrix A
$\det(A), \det A$	determinant of matrix A
$\text{Col}(A)$	column space of matrix A
$\text{Null}(A)$	null space of matrix A
$\text{Span}(\mathcal{A})$	linear span of the set of vectors $\mathcal{A}$
$\text{rank}(A)$	rank of a matrix A
$\dim(\mathcal{V})$	dimension of a vector space $\mathcal{V}$
$[\mathbf{x}]_{\mathcal{B}}$	the coordinate vector of $\mathbf{x}$ with respect to the basis $\mathcal{B}$

I Multiple Choices and True/False

Answer the following questions following the instructions.

Question 1 5 points

Consider two symmetric matrices A and B of the same size. Which of the following matrices are symmetric?

- (a) $A^2 - B^2$
- (b) $(A + B)(A - B)$
- (c) ABA
- (d) $ABAB$

Question 2 6 points

Consider the following matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}, \quad B = \begin{bmatrix} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & a_{21} \\ a_{34} & a_{33} & a_{32} & a_{31} \\ a_{44} & a_{43} & a_{42} & a_{41} \end{bmatrix},$$

$$P_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where A is invertible, and then $B^{-1} = (\text{_____})$

- (a) $A^{-1}P_1P_2$
- (b) $P_1A^{-1}P_2$
- (c) $P_1P_2A^{-1}$
- (d) $P_2A^{-1}P_1$

Question 3 6 points

Consider a matrix $A \in \mathbb{R}^{n \times n}$ and a vector $\mathbf{a} \in \mathbb{R}^{n \times 1}$, with

$$\text{rank} \begin{bmatrix} A & \mathbf{a} \\ \mathbf{a}^T & \mathbf{0} \end{bmatrix} = \text{rank}(A).$$

Then, (_____) is always true.

- (a) $A\mathbf{x} = \mathbf{a}$ has infinite number of solutions.
- (b) $A\mathbf{x} = \mathbf{a}$ has a unique solution.
- (c) $\begin{bmatrix} A & \mathbf{a} \\ \mathbf{a}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ y \end{bmatrix} = \mathbf{0}$ only has the trivial solution.

(d) $\begin{bmatrix} A & \mathbf{a} \\ \mathbf{a}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ y \end{bmatrix} = \mathbf{0}$ has a non-trivial solution.

Question 4 10 points

Determine whether each of the following statement is equivalent to that an $n \times n$ matrix A is nonsingular (invertible) by marking True or False. For the statement marked False, please correct the statement so that it is equivalent to A is nonsingular.

- (1) _____ $A\mathbf{x} = \mathbf{0}$ is consistent.
- (2) _____ $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b} \in \mathbb{R}^n$.
- (3) _____ A is row equivalent to I .
- (4) _____ There exists a matrix E such that $EA = I$.
- (5) _____ A is a product of a sequence of finite elementary matrices.
- (6) _____ The columns of A span $\mathbb{R}^n$.
- (7) _____ The rows of A form a basis of $\mathbb{R}^n$.
- (8) _____ A is full rank, i.e., $\text{rank}(A) = n$.
- (9) _____ $\dim(\text{Null}(A)) = 1$.
- (10) _____ $\det(A) > 0$.

II Calculations and Proofs

(Please give the details of your proofs and solutions. Your answer will be only judged based on your writing on the paper.)

Question 5 7 points

Let $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$. Calculate A^2 , A^3 and A^n . Suppose you know that

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Question 6 12 points

Let $A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$. Consider the following sequence of row operations:

$$A \xrightarrow{\text{op1}} \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ 0 & 0 & 0 & d-c \end{bmatrix} \xrightarrow{\text{op2}} \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix} \xrightarrow{\text{op3}} \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$$

- (a) (5 points) Specify the operation performed in each step of the row operations, and give the corresponding elementary matrices.
- (b) (3 points) Give an LU decomposition of A .
- (c) (4 points) Suppose that $a = 1$, $b = 2$, $c = 3$ and $d = 4$. Solve the system of linear equations: $A\mathbf{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$.

Question 7 10 points

Denote by P_4 the set of polynomial of degree less than 4. Let $\mathcal{A}$ be a subset of P_4 :

$$\mathcal{A} = \{1 + 2x^3, 2 + x - 3x^2, -x + 2x^2 + x^3, 1 + x - 3x^2 - 2x^3\},$$

where x is the indeterminate (variable) of the polynomials.

- (a) (5 points) Find the dimension of the subspace spanned by $\mathcal{A}$, together with a basis.
- (b) (5 points) Verify whether the polynomial $p_1 = 1 + 2x + 2x^2$ is in $\text{Span}(\mathcal{A})$ or not. If it is, give the coordinates with respect to the basis you find in (a).

Question 8 10 points

Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \mathbf{v}_1 = \begin{bmatrix} 4 \\ 6 \\ 7 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

- (a) (5 points) Find the change-of-basis transition matrix from $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.
- (b) (5 points) For $\mathbf{x} = 2\mathbf{v}_1 + 3\mathbf{v}_2 - 4\mathbf{v}_3$, determine the coordinates of $\mathbf{x}$ with respect to $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

Question 9 10 points

For the following matrix, find a basis for the row space, a basis for the column space and a basis for the null space.

$$\begin{bmatrix} 1 & 2 & -2 & 1 \\ 2 & 1 & 3 & 2 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$

Question 10 12 points

Consider the following linear system

$$\begin{cases} (a_1 + b)x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = 0 \\ a_1x_1 + (a_2 + b)x_2 + a_3x_3 + \cdots + a_nx_n = 0 \\ a_1x_1 + a_2x_2 + (a_3 + b)x_3 + \cdots + a_nx_n = 0 \\ \dots \\ a_1x_1 + a_2x_2 + a_3x_3 + \cdots + (a_n + b)x_n = 0 \end{cases},$$

where $\sum_{i=1}^n a_i \neq 0$, and a_i and b are some scalars. Questions:

- (6 points) When this linear system only has the trivial solution? (For which a_i and b ?)
- (6 points) When this linear system has non-trivial solutions? Then, obtain the solutions.

Question 11 12 points

- (4 points) Show that for $\mathbf{x}, \mathbf{y}_1, \dots, \mathbf{y}_n \in \mathbb{R}^m$ (where $m \geq n+1$), if $\mathbf{y}_1, \dots, \mathbf{y}_n$ are linearly independent and $\mathbf{x} \notin \text{Span}\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, then $\mathbf{x}, \mathbf{y}_1, \dots, \mathbf{y}_n$ are linearly independent.
- (8 points) Suppose $n \geq 3$ and $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}_3, \dots, \mathbf{d}_n \in \mathbb{R}^n$. Show that if $\mathbf{a} \notin \text{Span}\{\mathbf{b}, \mathbf{c}, \mathbf{d}_3, \dots, \mathbf{d}_n\}$, $\mathbf{b} \notin \text{Span}\{\mathbf{a}, \mathbf{c}, \mathbf{d}_3, \dots, \mathbf{d}_n\}$, and $\mathbf{c} \notin \text{Span}\{\mathbf{a}, \mathbf{b}, \mathbf{d}_3, \dots, \mathbf{d}_n\}$, then $\mathbf{d}_3, \dots, \mathbf{d}_n$ are linearly dependent.